Task 4: Short Question Answers:

Question 1: Rover position in grid coordinates

- Substract Rover's latitude and longitude from the origin. One degree of latitude is about 111 km, and one degree of longitude is about 111 km × cos(latitude). Over a 2×2 m area this flat-plane approximation is essentially exact, so a full map projection isn't needed.
- The grid is a few degrees off true north (angle θ) [Clue: grid's edges sit a few degrees off from the way the compass points.] Apply a 2D rotation by θ to the (east, north) offsets, which gives (x, y) along the grid's own axes.
- GPS reports where the rover is, not which way it faces, so position doesn't depend on heading. If heading is needed later, I would drive forward a short distance and take the direction between successive positions
- Ordinary GPS error is a few metres. I would use an RTK or differential GPS, or fuse GPS with wheel odometry and an IMU through a filter such as a Kalman filter. I would also confirm whether the offset angle is relative to true north or magnetic north, since mixing them introduces an error of several degrees.

Justification: The problem is a change of coordinate frame degrees → metres → rotated to match the grid's axes . Each step is simple, accurate at this scale, and independent of the rover's orientation.

Question 2: GPS dies when the IMU starts

USB-serial devices appear as /dev/ttyUSB0, /dev/ttyUSB1. Linux assigns these names in the order the devices are detected, so the names are not tied to a specific sensor. The IMU node was copied from the GPS node, so it likely still points at the same port, for example /dev/ttyUSB0. Two nodes then fight over one port, and the one that starts second disrupts the first. Both sensors work alone because each is the only device on that name. The failure flips after a reboot because the enumeration order changed, so the names swapped between the sensors.

Fix:

1. Create **udev rules** that give each device a permanent symlink, such as /dev/gps and /dev/imu. Match on attributes that are unique to each device, found with udevadm info -a -n /dev/ttyUSB0:

2. Update each node's config to use its own symlink, and check the baud rate and other settings that were copied from the GPS config.
3. Reload the rules with udevadm control --reload and udevadm trigger, then test by rebooting and replugging in different orders.
4. Make sure the user is in the dialout group so the ports are accessible.

Justification: Persistent names remove the dependence on boot or plug-in order, which was the root cause. Giving each sensor its own unique, fixed device path means the two nodes can never open the same port, so the problem cannot reappear.